Chapter 10: Correlation in R

Measuring Relationships Between Variables

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1 Overview

This reading prepares you for the lab session where you'll analyse **correlations** in R. Correlation is fundamental to psychology research — it tells us whether two variables are related and how strongly. You'll learn to compute correlation coefficients, test their significance, and create informative scatterplots.

> **ℹ What You'll Learn**
>
> By the end of this lab, you'll be able to:
>
> - Calculate correlation coefficients using `cor()` and `cor.test()`
> - Interpret correlation strength, direction, and significance
> - Create scatterplots with ggplot2 to visualise relationships
> - Understand the difference between correlation and causation

2 What Is Correlation?

Correlation measures the strength and direction of the linear relationship between two continuous variables.

- **Strength**: How tightly the data points cluster around a line (from weak to strong)
- **Direction**: Whether variables increase together (positive) or one increases as the other decreases (negative)

2.1 Correlation Coefficient (r)

The correlation coefficient **r** ranges from −1 to +1:

Value	Interpretation
r = +1	Perfect positive relationship
r = +0.7	Strong positive relationship
r = +0.3	Weak positive relationship
r = 0	No linear relationship

Value	Interpretation
r = −0.3	Weak negative relationship
r = −0.7	Strong negative relationship
r = −1	Perfect negative relationship

! Direction vs Strength

The **sign** tells you direction; the **absolute value** tells you strength. r = −0.8 is a stronger relationship than r = +0.3.

3 Computing Correlations in R

3.1 The `cor()` Function

For a quick correlation coefficient:

```
# Correlation between two variables
cor(datalab$reaction_time, datalab$time_estimate)
```

This returns just the correlation coefficient (e.g., 0.42).

3.2 The `cor.test()` Function

For a complete analysis including significance testing:

```
# Full correlation test
cor.test(datalab$reaction_time, datalab$time_estimate)
```

3.3 Understanding `cor.test()` Output

```
    Pearson's product-moment correlation

data:  datalab$reaction_time and datalab$time_estimate
t = 3.12, df = 48, p-value = 0.003
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.15  0.63
sample estimates:
      cor
0.42
```

Output	Meaning
`t = 3.12`	The t-statistic for testing if r differs from 0

Output	Meaning
`df = 48`	Degrees of freedom (n - 2)
`p-value = 0.003`	Probability of this r (or stronger) if true r = 0
`95 percent confidence interval`	Plausible range for the true correlation
`cor = 0.42`	The sample correlation coefficient

Interpreting the CI

A 95% CI of [0.15, 0.63] means we're fairly confident the true population correlation is somewhere in this range. Since it doesn't include 0, the correlation is statistically significant.

4 Correlation Matrix

When you have multiple variables, you can compute all pairwise correlations at once:

```
# Select numeric columns
numeric_vars <- datalab[, c("reaction_time", "time_estimate", "mood_pre",
"mood_post")]

# Correlation matrix
cor(numeric_vars)
```

This produces a matrix showing all correlations:

```
              reaction_time time_estimate mood_pre mood_post
reaction_time          1.00          0.42     0.15      0.18
time_estimate          0.42          1.00     0.08      0.12
mood_pre               0.15          0.08     1.00      0.65
mood_post              0.18          0.12     0.65      1.00
```

4.1 Handling Missing Values

If your data has missing values, add `use = "complete.obs"`:

```
cor(numeric_vars, use = "complete.obs")
```

5 Visualising Correlations with ggplot2

5.1 Basic Scatterplot

```
library(ggplot2)
```

```
ggplot(datalab, aes(x = reaction_time, y = time_estimate)) +
  geom_point()
```

5.2 Adding a Trend Line

```
ggplot(datalab, aes(x = reaction_time, y = time_estimate)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE)
```

- `method = "lm"` adds a linear regression line
- `se = TRUE` shows the confidence band around the line

5.3 Improved Scatterplot

```
ggplot(datalab, aes(x = reaction_time, y = time_estimate)) +
  geom_point(alpha = 0.6, size = 3) +
  geom_smooth(method = "lm", se = TRUE, color = "#275882") +
  labs(
    x = "Reaction Time (ms)",
    y = "Time Estimate (seconds)",
    title = "Relationship Between Reaction Time and Time Perception"
  ) +
  theme_minimal()
```

5.4 Scatterplot with Groups

To show different groups with different colours:

```
ggplot(datalab, aes(x = reaction_time, y = time_estimate, color = room)) +
  geom_point(alpha = 0.6, size = 3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    x = "Reaction Time (ms)",
    y = "Time Estimate (seconds)",
    color = "Room"
  ) +
  theme_minimal()
```

6 Types of Correlation

6.1 Pearson's r (default)

Assumes:

- Both variables are continuous
- Relationship is linear

- Data are approximately normally distributed

```r
cor.test(x, y, method = "pearson")  # default
```

6.2 Spearman's rho

Use when:

- Data are ordinal (ranked)
- Relationship is monotonic but not linear
- Data have outliers or are skewed

```r
cor.test(x, y, method = "spearman")
```

6.3 Kendall's tau

More robust alternative to Spearman's, good for small samples:

```r
cor.test(x, y, method = "kendall")
```

7 Key Functions Reference

Function	Purpose	Example
`cor()`	Quick correlation coefficient	`cor(x, y)`
`cor.test()`	Full test with p-value and CI	`cor.test(x, y)`
`cor(matrix)`	Correlation matrix	`cor(data[, vars])`
`geom_point()`	Add points to ggplot	`+ geom_point()`
`geom_smooth()`	Add trend line	`+ geom_smooth(method = "lm")`

7.1 Useful ggplot2 Arguments

Argument	Purpose	Example
`alpha`	Point transparency	`geom_point(alpha = 0.5)`
`size`	Point size	`geom_point(size = 3)`
`color`	Point/line colour	`geom_point(color = "blue")`
`se`	Show confidence band	`geom_smooth(se = TRUE)`
`method`	Smoothing method	`geom_smooth(method = "lm")`

8 Common Errors and How to Fix Them

8.1 Error: ‘x’ and ‘y’ must have the same length

```
# Wrong: different lengths
cor(datalab$reaction_time[1:40], datalab$time_estimate)
```

Fix: Ensure both variables have the same number of observations. This usually happens with missing values:

```
# Create complete cases
complete_data <- datalab[complete.cases(datalab$reaction_time,
datalab$time_estimate), ]
cor(complete_data$reaction_time, complete_data$time_estimate)
```

8.2 Error: ‘x’ must be numeric

Fix: Convert to numeric if needed:

```
# Check variable type
class(datalab$reaction_time)

# Convert if needed
datalab$reaction_time <- as.numeric(datalab$reaction_time)
```

8.3 Warning: Cannot compute exact p-value with ties

This happens with Spearman’s correlation when there are tied ranks.

Fix: This is usually fine — R provides an approximation. For small samples, consider Kendall’s tau instead.

8.4 The scatterplot looks empty

Fix: Check for issues with your data:

```
# Check for NAs
sum(is.na(datalab$reaction_time))

# Check for extreme values
summary(datalab$reaction_time)
```

8.5 Points are overplotting (too many on top of each other)

Fix: Use transparency or jitter:

```
# Transparency
geom_point(alpha = 0.3)
```

```r
# Or jitter (adds small random noise)
geom_jitter(width = 0.5, height = 0.5)
```

9 Correlation is NOT Causation

⚠ Critical Point

A correlation between X and Y does NOT mean X causes Y (or Y causes X). There are several possibilities:

1. X causes Y
2. Y causes X
3. A third variable (Z) causes both
4. The relationship is coincidental

Only controlled experiments can establish causation.

9.1 Classic Example

There's a strong correlation between ice cream sales and drowning deaths. Does ice cream cause drowning? No — a third variable (hot weather) increases both ice cream consumption and swimming.

10 Reporting Correlations (APA Style)

> Reaction time and time estimation were significantly positively correlated, r(48) = .42, p = .003, 95% CI [.15, .63].

Elements to include:

- The variables being correlated
- Direction (positive/negative)
- Degrees of freedom in parentheses
- The r value
- The p-value
- The 95% CI

11 Preparation Checklist

Before arriving at the lab:

⚠ Before You Arrive

- ☐ Read this chapter carefully
- ☐ Review the lecture material on correlation
- ☐ Understand what r values mean (strength and direction)
- ☐ Familiarise yourself with ggplot2 basics
- ☐ Have a device with a web browser ready (for WebR)

11.1 Questions to Consider

- Which DataLab variables might be correlated?
- What would a positive correlation between reaction time and time estimate mean?
- How would you check if a correlation is "real" versus due to chance?

11.2 What to Bring

- Laptop, tablet, or phone with a modern browser
- Ideas about relationships to test in the data
- Questions about interpreting correlations

12 Summary

Concept	Key Point
Correlation (r)	Measures strength and direction of linear relationship
Range	−1 (perfect negative) to +1 (perfect positive)
`cor()`	Quick coefficient only
`cor.test()`	Full test with p-value and CI
Scatterplot	Essential for visualising relationships
Causation	Correlation does NOT imply causation

13 Quick Reference

```
# Basic correlation
cor(x, y)

# With significance test
cor.test(x, y)

# Spearman (for non-normal data)
cor.test(x, y, method = "spearman")

# Correlation matrix
```

```r
cor(data[, c("var1", "var2", "var3")])

# Scatterplot with trend line
ggplot(data, aes(x = var1, y = var2)) +
  geom_point() +
  geom_smooth(method = "lm")
```

14 Additional Resources

- R Documentation for cor.test — Official documentation
- ggplot2 Reference: geom_point — Scatterplot options
- Correlation in R: Comprehensive Guide — Detailed tutorial